



Insights into emerging organic pollutants extraction from polypropylene, polystyrene, and polyethylene microplastics

Gabriel Jiménez-Skrzypek^{a,b}, Rachele Lusiardi^a, Javier González-Sálamo^{a,b,**}, Daura Vega-Moreno^c, Javier Hernández-Borges^{a,b,*}

^a Departamento de Química, Unidad Departamental de Química Analítica, Facultad de Ciencias, Universidad de La Laguna (ULL), Avda. Astrofísico Fco. Sánchez, s/n, 38206 San Cristóbal de La Laguna, Spain

^b Instituto Universitario de Enfermedades Tropicales y Salud Pública de Canarias, Universidad de La Laguna (ULL), Avda. Astrofísico Fco. Sánchez, s/n, 38206 San Cristóbal de La Laguna, Spain

^c Departamento de Química, Universidad de Las Palmas de Gran Canaria (ULPGC), Spain

HIGHLIGHTS

- Contaminants extraction from microplastics involve important challenges.
- Microplastics size, weathering and polymeric matrix effect should be studied.
- Microplastics cryomilling allows a suitable homogenization and replicate analysis.
- Smaller particle sizes highly improve the recoveries values obtained.
- Weathered microplastics show higher matrix effect than pristine ones.

ARTICLE INFO

Keywords:

Microplastics
Cryomilling
Contaminants
Sorption
Desorption

ABSTRACT

Background: Microplastics have the capability of retaining contaminants on their surface, increasing their persistence, preconcentrating them, and acting as transport vectors. Nevertheless, the determination of these compounds in plastic matrices poses several analytical issues and challenges, including the capability of many of these methods of only determining the extractable pollutants fractions, repeatability issues, etc. In this sense, it is primordial to evaluate the effect of the critical parameters that allow to obtain a quantitative extraction of the target analytes from microplastics, including the matrix effect of each of the studied polymers, the influence of particle size, and the effect of weathering.

Results: A simple and effective methodology for the extraction of 17 emerging organic pollutants from both pristine (polypropylene, polystyrene, and low- and high-density polyethylene) and weathered (polypropylene and polyethylene) microplastics has been developed, optimized, and validated, achieving recovery values of 70–120 % and low method quantification limits (9.2–35.5 ng/g). Results show the importance of cryomilling microplastics (as smaller particle sizes improve recovery and homogenization), something ignored in most publications. The differences in matrix effect for the studied pristine polymers highlights the importance of treating polymers individually, without extrapolating results. In weathered microplastics, matrix effect is overall higher than in their pristine counterparts, evidencing the necessity of always carrying out matrix effect and recovery studies in environmental microplastics. The analysis of 10 samples collected in Playa Grande (Tenerife, Canary Islands, Spain) revealed quantitative amounts of bisphenol A (10.8 ± 3.4 ng/g) in one of them.

Significance: For the first time, the effect of particle size, weathering and matrix effect have been simultaneously evaluated on microplastics, revealing the importance of their assessment to properly validate the methodology.

* Corresponding author. Departamento de Química, Unidad Departamental de Química Analítica, Facultad de Ciencias, Universidad de La Laguna (ULL), Avda. Astrofísico Fco. Sánchez, s/n, 38206 San Cristóbal de La Laguna, Spain.

** Corresponding author. Departamento de Química, Unidad Departamental de Química Analítica, Facultad de Ciencias, Universidad de La Laguna (ULL), Avda. Astrofísico Fco. Sánchez, s/n, 38206 San Cristóbal de La Laguna, Spain.

E-mail addresses: jgsalamo@ull.edu.es (J. González-Sálamo), jhborges@ull.edu.es (J. Hernández-Borges).

<https://doi.org/10.1016/j.aca.2023.342071>

Received 23 October 2023; Received in revised form 20 November 2023; Accepted 25 November 2023

Available online 2 December 2023

0003-2670/© 2023 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Additionally, the method shows good performance in all the different polymers and has been successfully applied to the analysis of environmental samples of microplastics.

1. Introduction

Every year, millions of tons of plastic are generated to meet the increasing global demand, reaching 400.3 million tons in 2022 [1]. However, the combination of exceptionally high production and inadequate plastic waste management, has led to the annual release of millions of tons of them into the environment [2]. As an example, it has even been reported that approximately up to 23 million tons of plastic are being released into the oceans annually [3,4].

Once in the environment, plastics are exposed to several degradation mechanisms (mechanical, chemical, or biological) [5]. Over time, these processes result in the fragmentation of plastic materials into smaller particles, categorised by size, shape, colour and composition. When these plastic particles range between 1 μm and 5 mm they are frequently classified as microplastics (MPs) -in particular, secondary MPs- [6]. However, it is important to note the fact that MPs are not only the byproduct of degradation; some are directly released into the environment with that size. This is the case of plastic pellets, microbeads or even glitter, which are called primary MPs [6]. Up to this date, published research has shown that MPs have been detected in all environmental compartments (i.e., water [7], air [8], and soils [9]), even in biota [10], indicating the widespread distribution and global transport of these environmental pollutants and their adverse effects at all levels.

For a long time, MPs have only been viewed as pollutants and, as such, they have been extracted from different matrices. Determining their amount, size, shape, colour and composition has posed significant challenges, especially with micron-size MPs [11,12]. Nevertheless, in recent years, their behaviour as matrices has gained widespread recognition. Essentially, MPs are polymeric particles with inherent sorption capabilities for different chemicals, encompassing both inorganic and organic ones (hydrophobic and/or hydrophilic) [13–15]. The retention of these compounds can take place through adsorption, the most extensively discussed mechanism in the literature [16], which is a surface phenomenon that involves different types of interactions (electrostatic, π - π stacking, van der Waals, etc.) or, alternatively, absorption, a bulk phenomenon that involves physical interactions or chemical reactions within the particle [17,18].

Several factors influence the sorption of MPs, including the chemical properties of pollutants, the physical and chemical properties of the polymers, and environmental conditions. The nature of MPs, influenced by factors like polymer type, size, and surface area, plays a significant role. The crystallinity of MPs, whether semi-crystalline or amorphous, impacts sorption efficiency, as amorphous polymers with larger free volumes enhance contaminant absorption [19]. The glass-transition temperature (T_g) of each polymer is pivotal, influencing the transition between rubbery and glassy states [17,19]. In this sense, in the rubbery state, polymer chains gain heightened flexibility, increased mobility, and larger free volumes, all of which favour absorption processes [20, 21]. Conversely, in the glassy state, polymer chains become more tightly intertwined, leading to more interactions between them and a denser structure, that results in reduced chain mobility and limited free-volume availability, thereby promoting adsorption [18,20]. Continual degradation (weathering) introduces surface alterations (i.e. functional groups containing oxygen, like hydroxyl, ketone or carboxylic acid groups, among others), affecting sorption and facilitating the retention of more hydrophilic compounds that otherwise would not be retained [5,15,21,22]. The specific surface area of MPs, influenced by their morphology, dimensions, and aggregation capacity, promotes adsorption interactions [18]. Environmental parameters (pH, salinity, temperature, and dissolved organic matter) and contaminant properties (hydrophobicity, pKa, molecular structures, etc.) are relevant as the

interplay of these factors is crucial for understanding MPs' sorption characteristics in diverse environments [16].

The development of new methodologies for the determination of pollutants retained by MPs has emerged as a crucial research area, due to the potential of MPs to transport contaminants [23–25], which could ultimately leach under specific conditions [26–29] back into the surrounding media. In this context, significant concerns have been raised regarding the impact of MPs on biomagnification and bioaccumulation processes, due to their capacity to introduce predominantly lipophilic pollutants in the food chain [21]. Assessing the presence of retained chemicals on different MPs becomes critical for the evaluation of the potential risk of environmental contamination and human health risks associated with their exposure [30]. Nevertheless, the development of these methodologies is still in its early stages, with a limited number of publications available. To date, several reviews and book chapters have discussed in depth the overall situation surrounding the determination of pollutants in MPs [12,14,30,31]. From these publications, it is evident that non-polar pollutants, particularly persistent organic pollutants (polychlorinated biphenyls -PCBs-, organochlorinated pesticides, etc.), predominate. This prevalence explains the widespread use of gas chromatography techniques in the determination of pollutants in MPs [12, 14,30,31]. Moreover, the application of traditional techniques (e.g., Soxhlet, maceration, etc.) and the necessity to apply fractioning clean-up steps is common [12,14,30,31]. Besides, many of the extraction methodologies that have been applied have several insufficiencies [12, 14,30,31]. On the one hand, many studies have performed a direct and single analysis of environmental MPs of sizes above 1 mm, making replicate analysis impossible due to the high variability of pollutants concentration among environmental MPs. Furthermore, many publications treat MPs as a unique matrix with homogeneous characteristics, often overlooking differences in the composition of each polymeric matrix (i.e. polyethylene, PE, polypropylene, PP, polyamide, etc.) or the impact of weathering, which leads to composition changes over time, extrapolating the validation done in a specific polymer type to many others without assessing the suitability of the method. Moreover, in most cases, matrix effect is not even studied, and it is common to find that only solvent calibration has been carried out. Regarding the extraction capability of the procedures, recovery values are frequently low, and quantitative extraction is not achieved often, with many publications only reporting the pollutants that the method is able to extract, risking underestimation. Finally, as previously mentioned, classical extraction techniques are still employed, with their many drawbacks, including high consumption of solvents as well as complex sample treatments caused by the complete dissolution of the MPs.

To the best of our knowledge, this article presents for the first time, an insightful evaluation of the numerous parameters influencing pollutants extraction from MPs, including parameters like polymer type, particle size, extraction solvent, time and volume, etc. In this sense, the use of a cryomilling system has been incorporated into the procedure to obtain plastic particles of different sizes, avoiding critical issues like plastic melting, obtaining reproducible milling results and allowing the development of replicate analysis. On the other hand, a parameter commonly overlooked as it is the degree of weathering in MPs, has been thoroughly studied, with regards to the difference between pristine and aged MPs. Additionally, a complete evaluation of the matrix effect for the different polymers studied has been conducted, emphasizing the clear differences existing among the polymers studied and underscoring the necessity to treat them as distinct matrices.

2. Materials and methods

2.1. Reagents and materials

Standards of caffeine (Sigma-Aldrich, 99.5 %, CAS 58-08-2), trimethoprim (Sigma-Aldrich, 99.9 %, CAS 738-70-5), cyclophosphamide (Sigma-Aldrich, 90.0 %, CAS 50-18-0), bisphenol F (Sigma-Aldrich, 99.1 %, CAS 620-92-8), cortisone (Sigma-Aldrich, 98.4 %, CAS 53-06-5), carbamazepine (Sigma-Aldrich, 99.0 %, CAS 298-46-4), bisphenol A (Sigma-Aldrich, 99.0 %, CAS 80-05-7), boldenone (Sigma-Aldrich, 99.5 %, CAS 846-48-0), 17 β -estradiol (Sigma-Aldrich, 96.1 %, CAS 50-28-2), testosterone (Sigma-Aldrich, 99.5 %, CAS 58-22-0), 17 α -ethinylestradiol (Sigma-Aldrich, 99.7 %, CAS 57-63-6), estrone (Sigma-Aldrich, 99.5 %, CAS 53-16-7), levonorgestrel (Sigma-Aldrich, 99.9 %, CAS 797-63-7), progesterone (Sigma-Aldrich, 99.7 %, CAS 57-83-0), 4-octylphenol (Sigma-Aldrich, 99.9 %, CAS 1806-26-4), 4-nonylphenol (Sigma-Aldrich, 99.9 %, CAS 104-40-5), and gemfibrozil (Dr. Ehrenstorfer, 99.5 %, CAS 25812-30-0). The solutions of the analytes were prepared in methanol (MeOH, Honeywell, LC-MS grade) and stored in the dark in the freezer at -18°C . Table S1 of the Supplementary Material shows the chemical structures and properties of the studied organic emerging pollutants.

Acetonitrile (ACN, Honeywell, LC-MS grade), ethanol (EtOH, Supelco, GC-ECD-FID grade), MeOH (VWR international, LC-MS grade), ethyl acetate (EtAc, Supelco, LC-MS grade) and acetone (Supelco, GC-MS grade) were used as extraction solvents. PP (CAS 9003-07-0), polystyrene (PS, CAS 9003-53-6) and low-density PE (LDPE, CAS 9002-88-4), and high-density PE (HDPE, CAS 9002-88-4) MP pellets (with diameters between 2 and 5 mm) were acquired from GoodFellow (United Kingdom).

Volumetric glassware was cleaned by a 24-h immersion in a solution of Nochromix (Godax Laboratories; Cabin John, MD) in sulfuric acid (95 % w/w, VWR International). Meanwhile, nonvolumetric glassware underwent a heat treatment process at 550°C for 4–5 h using a Carbolite CWF 11/13 oven, with a 13 L capacity and a maximum operating temperature of 1100°C .

A Sartorius analytical balance model 224i-1S (Goettingen, Germany) and a VWR International Eurolab Mega Star 3.0 R centrifuge were employed. For the extraction of the pollutants from the MPs samples, a VWR International Eurolab Rear Top vortex was utilized.

2.2. Environmental microplastic pellets sampling

In this work, weathered MP pellets of PE and PP were collected at Playa Grande beach (Tenerife, Canary Islands, Spain), which is considered a hot spot of MPs arrival in the region. Such MPs were used to develop recovery and matrix effect studies with weathered MPs. Besides, weathered PE MP pellets were also collected once a month from January 2022 to November 2022 in the same beach (the amount of PP and PS pellets collected was not enough to allow the same type of study). Pellets collection was carried out using stainless-steel tweezers and introducing them in glass amber vials of 40 mL previously calcined at 550°C for 4–5 h. Samples were kept in the dark before their final analysis.

2.3. Microplastic characterization and milling

2.3.1. Fourier-transform infrared spectroscopy

A Cary 630 Fourier-transform infrared (FTIR) spectrometer, coupled with a single reflection diamond Attenuated Total Reflectance (ATR) module (Agilent Technologies; California, USA) was employed for the purpose of ascertaining the composition of pristine and weathered MPs. The instrument featured a ZnSe beamsplitter and a 1.3 mm diameter thermoelectrically cooled deuterium triglycine sulphate detector. The data acquisition process involved the acquisition of 32 scans per spectrum, applying the Happ-Genzel apodization function, with a spectral resolution of 8 cm^{-1} , covering the wavenumber range from 650 to 4000

cm^{-1} . The Agilent MicroLab PC FTIR software, augmented with polymer libraries, was utilized for data acquisition and spectral identification.

X-ray powder diffraction analysis was conducted with an X'Pert PRO Multi-Purpose Diffractometer (PANalytical, Netherlands), employing the Bragg-Brentano geometry. Data was acquired using a Cu-K α radiation (wavelength, $\lambda = 1.54056\text{ \AA}$), spanning an angular range from 5.01° to 80.00° , with a total data collection duration of 20 min [32].

Differential scanning calorimetry (DSC) analyses were performed using a DSC 822 E instrument (Mettler Toledo, Belgium). The equipment was equipped with an autosampler, purge gas controller, and a temperature modulator, and exhibited a rapid response time of under 2.3 s, and good thermal reproducibility with variations of less than 0.1°C . Additionally, a wide temperature range, from -150 to 700°C (with thermal accuracy of approximately $\pm 0.2^{\circ}\text{C}$), and a resolution of $0.04\text{ }\mu\text{W}$ could be achieved. Samples used for the tests ranged from 17.4 to 24.5 mg, and the temperature was increased gradually from 0 to 240°C at a constant rate of 10°C per minute [32].

2.3.2. Microplastic pellets cryomilling and sieving

For pristine and weathered MP pellets homogenization, a Cryomill (Retsch, Germany) was employed. Samples were placed inside a stainless-steel reactor of 50 mL containing a stainless-steel ball of 2.5 cm diameter. First, the reactor was allowed to reach the operating temperature (-196°C) using the automatic function at a frequency of 5 s^{-1} , and then the particles were milled for 4 min (PP, LDPE, HDPE) or 45 s (PS), for a varying number of cycles at a frequency of 30 s^{-1} : 1, 4, 5 or 8, in the case of PS, PP, LDPE and HDPE, respectively, to obtain 250–500 μm sizes –the number of cycles/milling time can be reduced to obtain bigger particles–. Between cycles, a 50 s waiting/cooling period was introduced at a frequency of 5 s^{-1} . After the milling process, the resulting material was sieved using various meshes with sizes 2–5 mm, 1–2 mm, 0.5–1 mm, and 250–500 μm . The different MP size ranges were achieved by reducing the number of cycles to 2 for PP and LDPE, and 4 for HDPE and by reducing the milling time to 30 s (PS). Under these milling conditions, particles of different sizes are obtained, which can be then separated using the different sieves.

2.4. UHPLC-MS/MS conditions

The chromatographic separation and determination of the analytes was carried out using an ultra-high performance liquid chromatograph (1260 Infinity II) equipped with a quaternary pump (1260 Infinity II), an autosampler with temperature-controlled vials (1260 Infinity II), a column thermostat (1260 MCT), and an MS/MS detector (model G6470B) operated in dynamic multiple reaction monitoring (MRM) mode and equipped with an electrospray ionization (ESI) source, from Agilent Technologies.

The mobile phase components were ACN and Milli-Q water both containing $0.05\text{ }\%$ (v/v) NH_3 (LC-MS grade, $\geq 25\text{ }\%$ in water), at a flow rate of 0.4 mL/min . Separation was performed at 45°C using an Infinity Lab Poroshell HPH-C $_{18}$ column ($50 \times 2.1\text{ mm}$, $2.7\text{ }\mu\text{m}$) and an Infinity Lab Poroshell HPH-C $_{18}$ guard column ($5 \times 2.1\text{ mm}$, $2.7\text{ }\mu\text{m}$) from Agilent Technologies. The elution program started with an initial mobile phase composition of 95:5 water:ACN (v/v); then, the organic phase was increased to 50 % in 1 min and after that raised to 95 % in 2 min, maintaining that composition for 2 min before returning to the initial conditions in 1 min and keeping the conditions for 2 min before the next injection. The total run time of the analysis was 8 min, and the injection volume was $5\text{ }\mu\text{L}$. Regarding the ESI source parameters, the drying gas temperature was 210°C with a flow rate of 7 L/min . The nebulizer pressure was set at 25 psi and the sheath gas temperature was set at 400°C with a flow rate of 12 L/min . The capillary voltage was 3000 V for both positive and negative polarity, and the nozzle voltage was 0 V and -500 V , respectively. The m/z values for the quantifier and qualifier, as well as the collision energy and polarity used in the source for each of the analytes are shown in Table S2 of the Supplementary Material.

The UHPLC-MS/MS setup was operated with Agilent MassHunter Workstation Data Acquisition software (version 10.1, build 10.1.67). Additionally, Agilent MassHunter Workstation Qualitative Analysis (version 10.0, build 10.0.10305.0) facilitated chromatogram extraction, while Agilent MassHunter Workstation Quantitative Analysis (version 10.1, build 10.1.733.0) aided in integration and data extraction.

2.5. Sample spiking procedure

All samples were spiked 2 h prior to the analysis to ensure that enough time for the interaction between the pollutants and the MPs was provided. For this purpose, MPs were weighted on Petri dishes and the spiking solution (1 mg/L in MeOH) was added on top of them, covering the maximum amount of the samples surface with the solution. After the set spiking time, MPs were transferred into centrifuge tubes to carry out the extraction.

2.6. Extraction procedure

For the extraction procedure, 0.5 g of MP pellets powder of 250–500 μm size were weighted in a 15 mL glass centrifuge tube. Then, 5 mL of solvent (acetone for LDPE, HDPE, and PP, and ACN for PS) were added, the sample was vortexed for 5 min and then the tubes were centrifuged at 2500 rpm for 3 min. After that, the supernatant was filtered (polyvinylidene fluoride, PVDF, 0.20 μm , Whatman™) to eliminate any plastic particles or debris, and 3 mL were collected, and dried under a gentle nitrogen stream. Once dried, the extract was reconstituted in 1 mL of a mixture of water:ACN 70:30 (v/v), filtered (PVDF, 0.20 μm , Whatman™) and injected in the UHPLC-MS/MS system. Fig. 1 shows a scheme of the general extraction procedure.

3. Results and discussion

3.1. Microplastic pellets characterization

The four studied pristine polymers (LDPE, HDPE, PP and PS) were analysed by ATR-FTIR spectroscopy. Fig. S1 of the Supplementary Material shows the ATR-FTIR spectra of all of them. PP spectra shows clear bands associated with tension movements of the $-\text{CH}_3$ and $-\text{CH}_2$ bonds

2900 cm^{-1} , C–C bonds at 1350–1450 cm^{-1} , and bending bands of $-\text{CH}_3$ at 1000–1200 cm^{-1} [33], while PS spectra reveals three groups of bands related to tension movements of C–H (2800–3200 cm^{-1}) and C–C (1400–1600 cm^{-1}) bonds within the aromatic ring, and $-\text{CH}_2$ bending and aromatic tensions between 700 and 800 cm^{-1} [33]. LDPE spectra has three groups of bands related to tension movements of C–H (2850–2915 cm^{-1}) and C–C (1460 cm^{-1}) bonds and $-\text{CH}_2$ bending (750 cm^{-1}) [34]. Finally, HDPE spectra has three groups of bands related to tension movements of C–H (2850–2915 cm^{-1}) and C–C (1460 cm^{-1}) bonds and $-\text{CH}_2$ bending (715, 720 cm^{-1}) [34]. Differentiation of LDPE and HDPE spectra results from the splitting peaks seen at 715–720 cm^{-1} of HDPE [34]. This phenomenon is known as crystalline splitting, which results from the higher degree of crystallinity in HDPE [34]. Overall, the evaluation of the obtained ATR-FTIR spectra confirms that the different MPs matched the composition indicated by the manufacturer in the label.

Fig. S2 of the Supplementary Material shows the ATR-FTIR spectra of weathered PP and PE, the only two weathered plastics that could be collected for this study, compared to their pristine counterparts. In the case of weathered MPs, there are clear variations on the spectra associated with the presence of oxygen atoms in the structure of the polymer, which clearly suggest that they are indeed weathered MPs and that they have been in ocean for a long time. The spectral analysis of the weathered MPs in comparison to the pristine samples, reveals spectral changes particularly evident at 1100–1140 cm^{-1} (reflecting C–O bonds and/or alkanes), 1500–1800 cm^{-1} (associated with carboxylic acids, C=O ketones, C=C and/or C=O bonds), and 3000–3700 cm^{-1} (attributable to hydroxyl groups) [35]. The presence of these bands on MPs collected from the environment can be used as an indicator to evidence that the samples are weathered (even though, determining precisely the degree of weathering is extremely complex).

Additionally, the four selected types of pristine MPs, specifically PP, PS, LDPE, and HDPE pellets (3–5 mm), underwent further characterization employing XRD and DSC, as carried out in a previous work of our group (see Figs. S3 and S4 of the Supplementary Material) [32]. Upon analyzing the diffractograms, the semicrystalline structure of PP, LDPE, and HDPE was confirmed, as evidenced by the presence of clearly defined diffraction peaks within a continuous signal. In contrast, the diffractogram of PS unequivocally demonstrated the amorphous nature

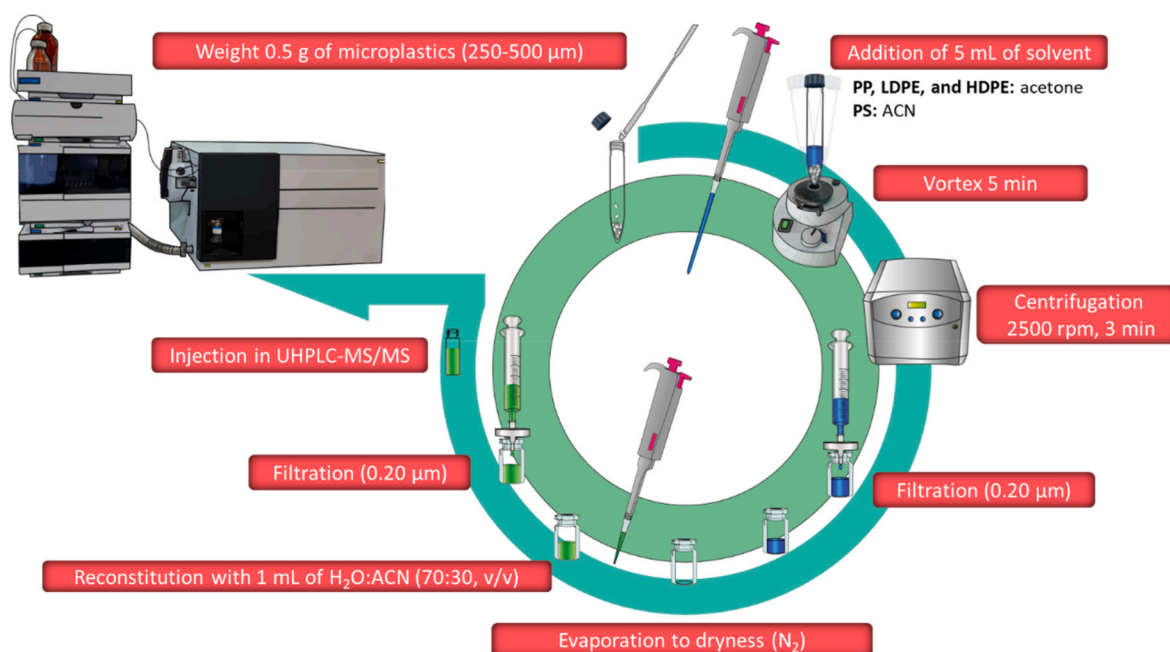


Fig. 1. General scheme of the vortex-assisted extraction-UHPLC-ESI-MS/MS method.

of this polymer, as it lacked distinct diffraction peaks [32]. On the other hand, the DSC measurements revealed important insights into the thermal behaviour of PP, LDPE, HDPE, and PS MPs. In this sense, PP, LDPE, and HDPE, exhibited endothermic peaks at approximately 165.7 °C, 112.1 °C, and 130.8 °C, respectively, signifying a melting process within these samples, which aligns with the semi-crystalline nature of these polymers, while in the case of PS, there was no exothermic peak associated with crystallization. Instead, an endothermic process occurred at 104.5 °C, corresponding to the T_g . During this transition, the amorphous polymer transforms from a glassy state with restricted chain mobility and reduced free volume to a rubbery state. In the rubbery state, chain mobility increases, and more free volume becomes available, further enhancing its potential for adsorption. Overall, the DSC analysis, reaffirms the amorphous nature of PS and the semi-crystalline structure of PP, LDPE, and HDPE, consistent with the XRD results [32].

3.2. UHPLC-MS/MS figures of merit

The determination of the 17 emerging pollutants was carried out through UHPLC-MS/MS (see Experimental Section for details) using an Infinity Lab Poroshell HPH-C₁₈ column (50 × 2.1 mm, 2.7 µm) resistant to high pH values and working in gradient mode with a mobile phase consisting of water and ACN (both containing 0.05 % ammonia), at 45 °C and a flowrate of 0.4 mL/min. The group of analytes selected included phenols (bisphenol A, bisphenol F, 4-octylphenol, and 4-nonylphenol), hormones (17 α -ethinylestradiol, 17 β -estradiol, boldenone, cortisone, estrone, levonorgestrel, progesterone, and testosterone), pharmaceuticals (carbamazepine, cyclophosphamide, gemfibrozil, and trimethoprim) and a consumption habit compound also used as stimulant in pharmaceutical formulations (caffeine), with few of which having been previously found in MPs [12,14,30,31].

To assess the precision of injection within the chromatographic system, a study was conducted to evaluate intra-day and inter-day repeatability. For this purpose, a solvent mixture containing the analytes was injected ($n = 5$) over three consecutive days at three distinct concentration levels (100, 250, and 400 ng/g). Tables S3 and S4 of the Supplementary Material present the RSD values obtained for peak areas and retention times, respectively, for each of the analysed compounds. Generally, the RSDs were ≤ 0.61 % for retention times and ≤ 13.9 % for peak areas within the same day. These values changed to ≤ 0.97 % for retention times and ≤ 15.3 % for peak areas when comparing results between different days. To assess the linearity of the method, a calibration curve of the different analytes at 10 concentration levels was obtained. Table S5 of the Supplementary Material shows the solvent calibration data of the target analytes. Due to the good signal-to-noise (S/N) ratios obtained in the UHPLC-MS/MS system, a lowest calibration level (LCL) which yielded a signal-to-noise ratio higher than 10 was used during instrumental calibration. Overall, the LCL for the different analytes in the instrumental calibration was established at 3 µg/L while the studied linear range was between 3 and 150 µg/L, with determination coefficients (R^2) for all analytes higher than 0.990. Fig. S5 of the Supplementary Material shows the MRM chromatogram obtained for the different analytes at 100 µg/L in solvent, showing a suitable chromatographic separation for the different analytes, with symmetric and well-defined peaks, being the analysis time less than 6 min.

3.3. Method optimization and validation

A step-by-step optimization of the method was carried out in the four pristine MPs selected (LDPE, HDPE, PP and PS), focusing on the different parameters influencing the extraction of the pollutants such as extraction solvent, vortex time, and extraction volume. Prior to the optimization of the method, a screening study using different particle sizes was conducted. From this initial study, comminuted MPs of 250–500 µm size were selected, since they reported the better results for recovery studies,

as it will be shown in section 3.3.4, in which the effect of particle size will be discussed in more detail.

MPs were crushed in a cryomill working at extremely low temperatures (-196 °C), favouring plastics milling whilst avoiding plastic melting, a critical step to take into consideration. Additionally, low temperatures facilitate plastic fragmentation, especially in the case of plastics such as HDPE, that due to their inherent characteristics are very resistant. Then, samples (0.5 g; 250–500 µm) were spiked 2 h prior to the analysis to ensure proper interaction between the pollutants and the MPs.

3.3.1. Effect of extraction solvent

In the case of the extraction solvent, several options were studied, including some of the most used solvents of different polarity such as ACN, MeOH, EtOH, acetone and EtAc; however, in the case of PS, acetone and EtAc had to be discarded as they dissolved the polymer. The initial conditions selected were: extraction of 0.5 g of each MP of 250–500 µm particle size, 5 min of vortex, and 4 mL of extraction solvent. All extractions were carried out in triplicate at a concentration of 400 ng/g. The selection of these solvents was made considering the Hildebrand and Hansen solubility parameters for each of the different polymers in different types of solvents [36]. Overall, the use of non-polar solvents with strong dissolving capability (e.g., cyclohexane, hexane, etc.) was avoided, as it may lead to negative effects on the chromatographic system and on matrix effect, and also as a result of the polarity of the selected analytes. In some other cases, where the solvent had slight dissolving capability, a compromise was made (e.g., acetone, EtAc, and ACN), since, due to the short extraction times, plastics dissolution would be greatly atoned. For each solvent and MP types, absolute recovery values were calculated using matrix-matched standards at the expected final concentration.

As it can be seen in Fig. 2, the recovery values obtained for each solvent were high in most cases. The evaluation of the effect of the different solvents in PP, showed that acetone and EtOH provided the best performance for most of the analytes (100 % of the analytes with recovery percentages ranging from 60 to 120 %), nevertheless, EtOH showed overall lower recovery for 4-octylphenol and 4-nonylphenol (60–63 %), which led to the selection of acetone as the preferred option. In the case of PS, the three solvents studied exhibited a similar behaviour, nonetheless, recovery for 4-octylphenol and 4-nonylphenol, were slightly higher in ACN (62.2–62.5 %), making this solvent as the optimal choice. LDPE and HDPE polymers showed the best performance with acetone (95–100 % of the analytes in the 60–120 % recovery range), with only HDPE showing lower recovery values for two of the analytes, 4-octylphenol and 4-nonylphenol (55.8 and 61.3 %, respectively).

After a thorough assessment of the results, it was decided that the best solvent for PP, LDPE, and HDPE MPs was acetone (95–100 % of the analytes showed recovery values between 60 and 120 %, with recovery values higher than the other solvents), whereas for PS, (acetone and EtAc could not be used) ACN was chosen, as it showed slightly higher recovery percentages for the analytes than MeOH and EtOH, especially for 4-octylphenol.

3.3.2. Effect of extraction solvent volume

After determining the optimal extraction solvent for the polymers, different extraction volumes were tested between 3 and 8 mL (acetone for LDPE, HDPE and PP and ACN for PS). For this purpose, 0.5 g of each MP of 250–500 µm particle size and 5 min of vortex were applied. Solvent volume below 2 mL was not useable since solvent collection was not possible. As shown in Fig. 3, overall recovery obtained for each of the different volumes were high, with differences among them being, in most cases, very subtle and with clear signs that the use of volumes higher than 6 mL did not report a significant improvement in the recovery values, for most analytes. In the case of PP, after evaluating the results, the best option was deemed to be 5 mL. The same volume was

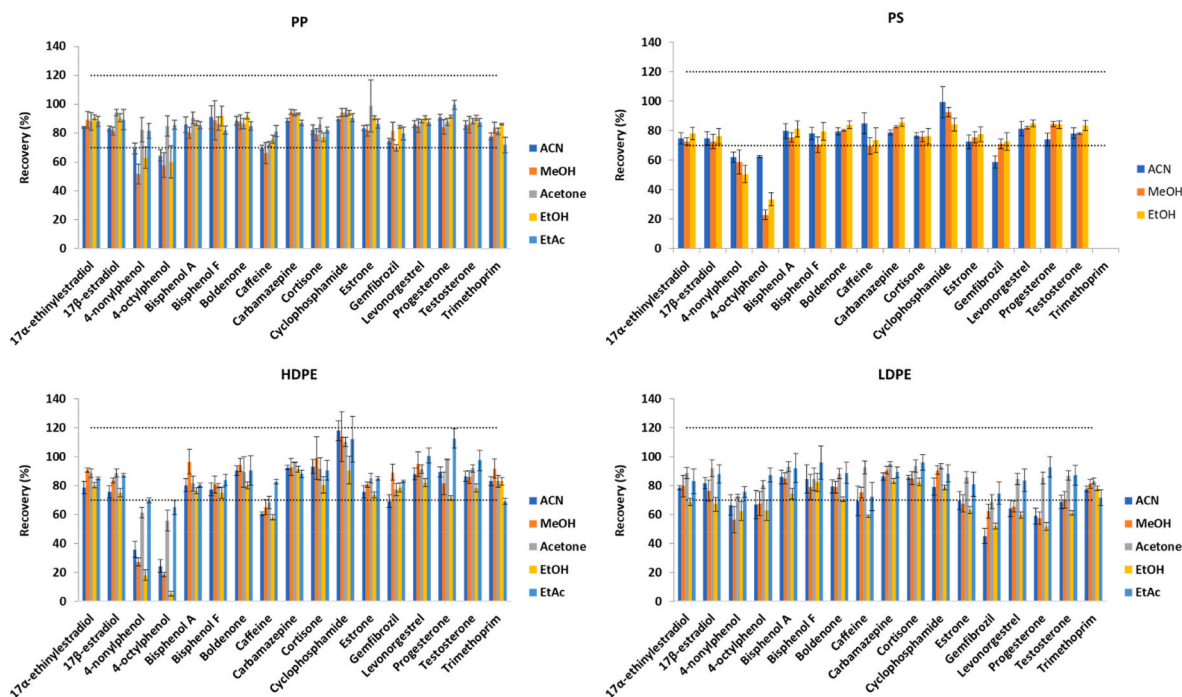


Fig. 2. Effect of the extraction solvent in the absolute recovery values of the target analytes. Extraction conditions: 0.5 g of each MP of 250–500 μm particle size, 5 min of vortex, and 4 mL of extraction solvent. Concentration of the target analytes in the MPs: 400 ng/g.

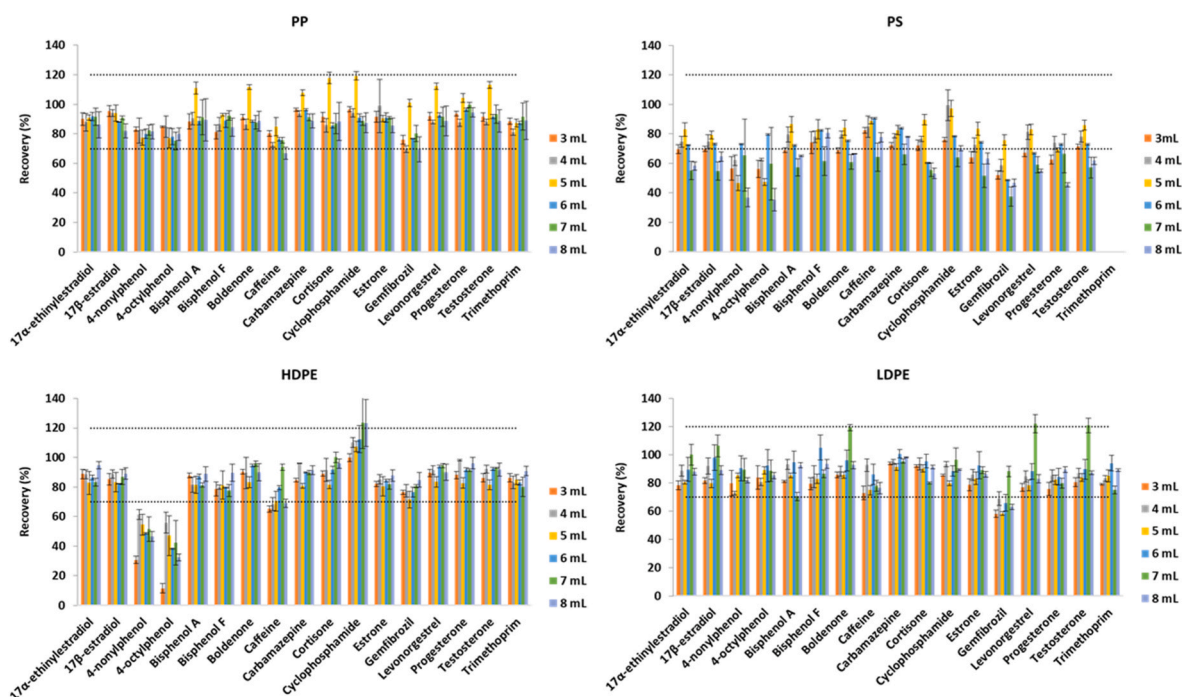


Fig. 3. Effect of the extraction solvent volume on the absolute recovery values of the target analytes. Extraction conditions: 0.5 g of each MP of 250–500 μm particle size, 5 min of vortex, and acetone for LDPE, HDPE and PP and ACN for PS. Concentration of the target analytes in the MPs: 400 ng/g.

also chosen for PS, as it provided the highest recovery for most analytes apart from 4-octylphenol and 4-nonylphenol, which showed low recovery values (47.5 % and 45.8 %, respectively). This low recovery tendency was observed for most extraction volumes, except for 6 mL; nevertheless, in that case, recovery values were overall lower for all the other analytes, and compounds like gemfibrozil showed also poor performance (48.6 % recovery). In the case of LDPE recovery percentages, they exhibited very small differences with the different volumes, except

for gemfibrozil, which had slightly lower recovery (58–68 %) except for 7 mL (88 % recovery). HDPE on the other hand, had a similar tendency to the one described for LDPE, with the difference that lower recovery values were observed for 4-octylphenol and 4-nonylphenol (recovery <62 %). Overall, LDPE and HDPE optimal extraction volumes were set between 4 and 6 mL, with no significant improvements being observed at higher volumes.

Given that the differences between 4 and 6 mL were not remarkably

high, and to homogenise the extraction conditions of the different polymers, a compromise was made, and 5 mL was selected as the extraction volume for all the polymers, as it provided good recovery and avoided the unnecessary consumption of solvents. Such volume was also selected in an attempt to obtain a similar extraction procedure for all MPs types and, therefore, to facilitate routine analysis.

3.3.3. Effect of extraction time

With all the previous parameters set, the effect of the extraction time was also studied. In this sense, extractions between 1 and 5 min were evaluated, and an additional time of 10 min was also included to assess if prolonged exposure to the solvent would result in a significant increase of the recovery. For these studies, 0.5 g of each MP of 250–500 μm particle size and 5 mL of acetone for LDPE, HDPE and PP and 5 mL of ACN for PS were applied.

For instance, in the case of PP, no significant differences were observed between the different times, as 95 % of the analytes showed recovery values between 60 and 120 %. For PS the best option was 5 min, as 88.9 % of the analytes had recovery percentages between 60 and 120 %, with 11.1 % falling in the 40–60 % range (4-octylphenol and 4-nonylphenol). In the case of LDPE, the best option was 5 min, as all the analytes exhibited recovery values in the 60–120 % range (with the lower recovery being observed for gemfibrozil; 68.8 %) whereas HDPE did not show a clear tendency, with good results being observed across the different extraction times. Nevertheless, to homogenise the experimental conditions, and since the differences between 4, 5, and 10 min were small, it was decided to choose 5 min as extraction time. Fig. S6 of the Supplementary Material shows the recovery values of the different analytes at the different extraction times.

Overall, the optimal extraction time was set at 5 min for all the polymers, as longer extraction times (e.g., 10 min) did not improve the recovery, and the results obtained for extraction time were good enough across the different polymers.

3.3.4. Evaluation of particle size effect

As it has already been mentioned at the beginning of this section, prior to the extraction optimization, several screening studies were performed to evaluate the effect of particle size on recovery, showing that small sizes reported better results. With the optimized method, a complete study was carried out to confirm the previously mentioned tendency overserved in the screenings. In this sense, MPs were cryomilled and sieved as previously described. The particles were separated in different sizes (250–500 μm , 0.5–1 mm, 1–2 mm, and 2–5 mm), and the optimized extraction method was applied to each type of particle size and for each type of polymer. Generally, results showed a clear tendency for most of the analytes, with an increase in recovery as

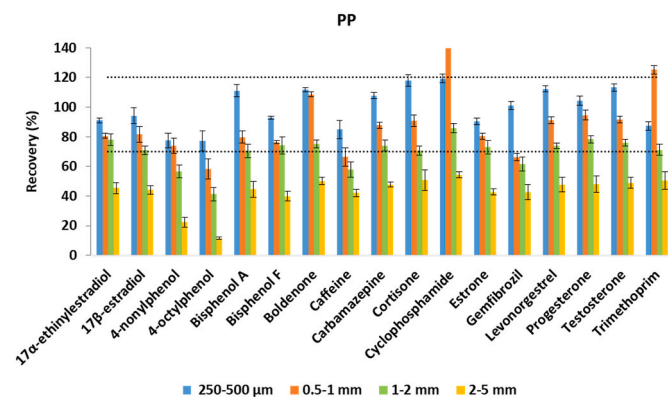


Fig. 4. Effect of the particle size of PP MPs in the absolute recovery values of the target analytes. Extraction conditions: 0.5 g of PP MPs, 5 mL of acetone as extraction solvent and 5 min of vortex. Concentration of the target analytes in the MPs: 400 ng/g.

particle size was reduced. Fig. 4 shows the recovery values obtained at 400 ng/g for the different analytes at each range of particle sizes for PP (Fig. S7 of the Supplementary Material contains the results obtained for each one of the different polymers). In this figure, it can be clearly seen the tendency afore mentioned, which is especially clear when we compare the results for 2–5 mm MP with those observed for the 250–500 μm particles, where the recovery values are almost the double of those observed in the bigger sizes. This difference is reduced, when particles are milled to a range of 0.5–1 mm; nevertheless, the recovery is still higher in the case of the 250–500 μm particles for most of the analytes. As it can be seen in the case of PP and HDPE, the difference between both types of sizes (2–5 mm and 250–500 μm) is particularly higher than in the cases of LDPE and PS.

From these results, it is clear that particle size plays a major role in extraction performance. Overall, smaller sizes report better results (something that could have been expected, but that has been experimentally proven). Additionally, the use of smaller particle sizes favours homogenization, allowing the development of replicate analysis without observing significant variations between samples. As for the use of smaller particle sizes (e.g., <250 μm), in this study it was chosen not to do so for two main reasons: first, because absolute recovery values for most of the analytes were already high (milling MPs to that size would require higher consumption of resources and time), and, second, to avoid several difficulties that could arise from the use of smaller particles during (manipulation problems, polymer dissolution, etc.).

As it has already been mentioned, many works in the literature only analyse millimetre size MPs [12,14,30,31] and, as a result, replicate analysis have provided an extremely high variability in the results, since the concentration of analytes in “non-homogenized” samples is high. Comminution of the MPs down to 250–500 μm will provide on the one hand, quantitative recovery values and, on the other, suitable and representative matrix for replicate analysis.

3.3.5. Recovery studies in pristine and weathered microplastics

After properly optimizing the method, recovery studies at three levels of concentration (100, 250 and 400 ng/g) were performed for each of the pristine MPs (LDPE, HDPE, PP and PS with sizes in the range 250–500 μm). The levels were chosen trying to cover a wide concentration range (given that there is reduced information regarding the presence of the studied pollutants in MPs) and trying to cover the whole calibration range (final concentration values). As depicted in Table 1, absolute recovery percentages were in the 71.4–103 % range with RSDs lower than 25 %, except for 4-octylphenol and 4-nonylphenol in PS and HDPE, which exhibited lower recovery values 37.9–46.9 % and 57.5–58.9 %, respectively. Additionally, a similar situation was observed for gemfibrozil in both of the polymers previously mentioned, exhibiting recovery values of 54.7 % in HDPE and 62.8 % in PS. Finally, in the case of HDPE, caffeine also exhibited slightly lower recovery than the average, with a mean recovery of 69.1 %. It should be noted that in the case of PS, trimethoprim recovery values could not be included, since they were inconsistent with an extremely high RSD value.

Similarly to the case of pristine MPs, a recovery study at three levels of concentration (100, 250 and 400 ng/g) was carried out on weathered MP pellets of PE and PP to evaluate the performance of the method. As previously indicated, this recovery study could only be carried out with weathered PP and PE, since PS pellets could not be recovered from Canary beaches in an enough quantity to develop such recovery study. Results are also shown in Table 1. Overall, absolute recovery in both weathered polymeric matrices ranged from 72.1 to 109 % with RSDs below 16 %. In the case of weathered PE, all analytes showed recovery values between 70 and 120 % (with slightly lower mean recovery values being observed for caffeine, 72.1 %, and trimethoprim, 76.3 %), while in the case of weathered PP, most analytes were also in the 70–120 % range, with compounds like trimethoprim showing lower values (60.8 %). These results clearly indicate that recovery studies are consistent with weathered PP and PE for the selected target analytes, as the

Table 1
Absolute recovery and RSD (%) values of the target analytes in the different polymers studied.

	Recovery values for PP				Recovery values for LDPE				Recovery values for HDPE				Recovery values for PS				Recovery values for weathered PP				Recovery values for weathered PE			
	100	200	400	Mean	100	200	400	Mean	100	200	400	Mean	100	200	400	Mean	100	200	400	Mean	100	200	400	Mean
	ng/g	ng/g	ng/g		ng/g	ng/g	ng/g		ng/g	ng/g	ng/g		ng/g	ng/g	ng/g		ng/g	ng/g	ng/g		ng/g	ng/g	ng/g	
17α-ethinylestradiol	83.8	87.0	91.0	87.3	78.2	79.7	80.5	79.5	93.0	77.0	82.9	84.3	67.8	66.4	83.0	72.4	88.4	80.0	87.3	85.2	93.9	93.8	94.9	94.2
17β-estradiol	(1.2)	(2.2)	(1.7)	(4.1)	(2.4)	(1.9)	(1.2)	(1.5)	(0.9)	(4.2)	(7.8)	(9.6)	(2.6)	(7.2)	(4.6)	(13)	(5.8)	(3.5)	(1.3)	(5.3)	(7.5)	(2.4)	(1.5)	(0.7)
4-nonylphenol	85.1	86.7	94.1	88.6	79.0	81.8	79.6	80.1	88.7	79.9	83.3	84.0	71.0	71.4	79.3	73.9	90.9	77.8	86.7	85.2	92.3	97.0	92.2	93.9
	(1.8)	(1.6)	(5.3)	(5.4)	(1.7)	(4.0)	(2.8)	(1.9)	(1.6)	(2.3)	(6.5)	(5.3)	(1.9)	(8.3)	(2.6)	(6.3)	(3.0)	(3.6)	(3.4)	(7.9)	(4.3)	(2.0)	(2.5)	(2.9)
4-octylphenol	92.2	68.5	77.5	79.4	85.2	57.8	85.5	76.2	65.1	50.4	57.1	57.5	54.1	40.7	45.8	46.9	91.7	74.9	83.1	83.2	72.3	81.0	78.7	77.3
	(2.7)	(2.4)	(5.0)	(15)	(5.9)	(4.1)	(1.7)	(21)	(9.3)	(11)	(7.7)	(13)	(7.6)	(5.1)	(14)	(14)	(7.5)	(5.1)	(4.2)	(10)	(5.4)	(3.3)	(6.6)	(5.9)
Bisphenol A	73.0	65.5	77.3	71.9	79.8	57.8	89.0	75.5	68.5	55.9	52.3	58.9	34.0	35.1	47.5	38.9	81.1	65.9	80.8	75.9	77.7	86.0	85.5	83.1
	(3.7)	(2.3)	(6.8)	(8.3)	(4.7)	(6.0)	(2.6)	(21)	(10)	(10)	(14)	(14)	(6.4)	(0.2)	(4.9)	(19)	(5.7)	(11)	(3.1)	(11)	(3.1)	(5.3)	(2.4)	(5.6)
Bisphenol F	83.5	83.9	111	92.8	82.6	83.8	85.4	83.9	92.5	73.9	81.4	82.6	68.7	68.6	86.6	74.6	96.5	83.6	87.4	89.2	94.6	97.2	94.9	95.6
	(2.9)	(1.9)	(4.1)	(17)	(5.1)	(1.3)	(1.7)	(1.7)	(4.5)	(3.6)	(5.7)	(11)	(6.2)	(8.8)	(5.4)	(14)	(5.9)	(4.4)	(3.6)	(7.4)	(4.9)	(2.4)	(3.0)	(1.5)
Boldenone	83.6	83.8	92.9	86.8	85.5	91.9	82.9	86.8	87.7	73.2	81.2	80.7	74.8	72.6	83.2	76.9	90.1	74.9	84.7	83.3	104	95.5	94.7	98.1
	(1.4)	(1.8)	(0.9)	(6.1)	(1.7)	(4.6)	(3.2)	(5.4)	(4.2)	(4.3)	(9.5)	(9.0)	(3.7)	(12)	(6.6)	(7.3)	(8.3)	(7.3)	(7.0)	(9.3)	(2.2)	(7.1)	(5.9)	(5.2)
Caffeine	92.5	80.0	112	94.7	80.3	82.2	85.7	82.7	85.0	76.8	83.3	81.7	67.5	73.6	84.1	75.1	97.7	80.5	86.1	88.1	96.7	97.0	93.1	95.6
	(0.3)	(0.4)	(1.6)	(17)	(2.2)	(2.7)	(2.2)	(3.3)	(2.4)	(0.9)	(3.7)	(5.3)	(1.0)	(4.6)	(5.1)	(11)	(8.4)	(3.4)	(6.3)	(9.9)	(5.1)	(2.4)	(1.9)	(2.3)
Carbamazepine	74.6	65.2	85.0	74.9	74.4	72.2	74.9	73.8	69.6	67.5	70.4	69.1	78.6	70.3	88.9	79.2	88.3	64.3	74.7	75.8	68.2	71.9	76.1	72.1
	(3.8)	(3.2)	(6.1)	(13)	(2.8)	(1.7)	(3.4)	(1.9)	(1.9)	(3.6)	(6.8)	(2.2)	(5.1)	(7.9)	(2.1)	(12)	(1.2)	(1.9)	(6.5)	(16)	(4.3)	(1.3)	(0.9)	(5.5)
Cortisone	95.9	86.5	108	96.7	82.9	88.1	91.4	87.5	86.6	73.9	80.8	80.5	71.6	71.8	82.5	75.3	98.8	80.2	85.5	88.2	94.9	93.3	94.2	94.1
	(1.8)	(2.2)	(2.1)	(11)	(1.1)	(4.3)	(2.4)	(4.9)	(5.3)	(1.5)	(1.5)	(7.9)	(1.2)	(4.9)	(3.2)	(8.3)	(5.0)	(1.6)	(6.3)	(11)	(6.0)	(1.6)	(1.7)	(0.8)
Cyclophosphamide	71.9	87.8	118	92.5	88.0	91.4	90.2	89.9	86.9	77.5	81.6	82.0	70.0	69.8	89.7	76.5	82.1	74.6	82.9	79.9	87.8	91.9	91.8	90.5
	(1.3)	(4.9)	(3.8)	(25)	(5.4)	(1.4)	(2.6)	(1.9)	(5.1)	(0.9)	(2.8)	(5.8)	(1.0)	(6.2)	(3.5)	(15)	(13)	(5.3)	(9.8)	(5.7)	(4.4)	(2.8)	(1.2)	(2.6)
Estrone	97.2	91.4	119	103	90.3	101	79.9	90.5	107	91.2	107	102	79.8	75.7	97.2	84.2	97.6	77.6	87.0	87.4	111	102	113	109
	(2.3)	(7.5)	(2.7)	(14)	(1.9)	(13)	(1.6)	(12)	(11)	(3.0)	(3.6)	(9.1)	(1.1)	(6.8)	(5.4)	(14)	(3.0)	(3.7)	(6.4)	(12)	(7.2)	(4.3)	(0.2)	(5.2)
Gemfibrozil	86.7	84.8	90.4	87.3	76.4	74.2	83.2	77.9	83.9	77.0	80.4	80.4	68.4	66.5	83.4	72.8	88.5	78.6	88.3	85.1	100	97.4	92.7	96.7
	(2.1)	(1.4)	(2.1)	(3.3)	(2.4)	(0.6)	(4.4)	(6.0)	(1.9)	(3.2)	(6.5)	(4.3)	(4.9)	(6.4)	(4.5)	(13)	(0.9)	(4.4)	(2.3)	(6.6)	(6.5)	(2.0)	(1.5)	(3.9)
Levonorgestrel	74.3	64.8	101	80.0	55.7	50.3	58.2	54.7	73.8	68.8	71.4	71.4	55.1	57.7	75.8	62.8	94.5	73.3	84.6	84.1	95.8	99.7	90.4	95.3
	(4.5)	(0.3)	(2.5)	(23)	(1.1)	(1.5)	(1.3)	(7.4)	(3.9)	(1.8)	(5.7)	(3.5)	(2.3)	(4.9)	(3.5)	(18)	(0.5)	(4.1)	(4.8)	(13)	(4.0)	(0.8)	(0.8)	(4.9)
Progesterone	93.5	86.8	112	97.5	77.4	73.5	78.4	76.4	89.6	78.7	83.5	83.9	75.9	69.4	83.0	76.1	98.0	79.3	85.4	87.6	98.3	98.7	88.8	95.2
	(0.6)	(0.9)	(2.2)	(14)	(1.1)	(3.2)	(3.9)	(3.4)	(4.1)	(1.1)	(3.6)	(6.5)	(4.5)	(2.5)	(3.6)	(8.9)	(4.2)	(2.6)	(8.7)	(11)	(4.9)	(2.3)	(2.6)	(5.9)
Testosterone	95.0	89.3	104	96.2	71.2	72.0	82.2	75.2	86.0	79.3	82.7	82.6	77.8	67.9	69.2	71.6	103	82.3	86.4	90.5	97.3	99.6	90.8	95.9
	(0.7)	(1.4)	(3.2)	(7.9)	(0.9)	(1.0)	(3.3)	(8.2)	(3.2)	(2.7)	(3.7)	(4.1)	(5.2)	(2.3)	(1.5)	(7.5)	(3.6)	(3.4)	(7.8)	(12)	(4.2)	(1.1)	(1.4)	(4.8)
Trimethoprim	92.7	82.8	108	94.5	75.9	75.4	83.8	78.3	88.5	82.1	81.4	84.0	75.2	71.7	86.0	77.6	97.0	80.1	87.5	88.2	101	98.9	90.7	96.8
	(1.3)	(0.7)	(2.2)	(14)	(0.7)	(1.1)	(2.8)	(6.0)	(0.6)	(1.1)	(3.4)	(4.7)	(1.2)	(5.8)	(3.4)	(9.6)	(5.2)	(2.1)	(7.5)	(9.6)	(3.6)	(1.1)	(2.6)	(5.5)
	88.5	71.2	87.4	82.4	78.7	79.3	85.0	81.0	86.0	82.0	84.9	84.3	–	–	–	–	65.3	59.6	57.4	60.8	65.8	76.0	86.9	76.3
	(1.2)	(4.3)	(2.8)	(12)	(4.2)	(1.1)	(4.1)	(4.3)	(2.1)	(1.0)	(3.3)	(2.5)					(4.5)	(8.9)	(9.8)	(6.7)	(21)	(15)	(0.4)	(14)

analytes can be highly recovered with good RSDs both from weathered and pristine MPs, despite the higher matrix effect that was observed in weathered MPs, and which will be discussed in section 3.3.6.

3.3.6. Matrix-matched calibration studies and matrix effect assessment

To evaluate the linearity of the method, matrix-matched calibration was developed for all the analytes in each polymeric matrix (including pristine and weathered polymers) at ten increasing levels of concentration ($n = 10$) injecting each concentration in triplicate. Overall, the studied linear ranges were between 3 and 150 $\mu\text{g/L}$ for most analytes except for those studied in PS where the linear range was from 3 to 125 $\mu\text{g/L}$ or 5–125 $\mu\text{g/L}$, and cortisone in weathered PP, where the range was from 10 to 150 $\mu\text{g/L}$. In all cases R^2 were higher than 0.991, proving the

good linearity of the method (see Table S5 of the Supplementary Material). As a result of the good absolute recovery values obtained for most of the analytes, low LOQs of the method were obtained (9.2–35.5 ng/g), which were calculated from the LCL taking into consideration the absolute recovery values obtained for each analyte and polymeric matrix. These values were also experimentally checked by carrying out extractions at similar concentrations. Regarding the LOQs of the method, Table S6 of the Supplementary Material compiles the results obtained for each one of the different analytes.

Due to the differences intrinsic to each type of matrix, an evaluation of the matrix effect was done both in the pristine MPs (PP, LDPE, HDPE, and PS) as well as the weathered ones (PP and PE). To do so, the Matuszewski et al. [37] equation was used. In this sense, the matrix

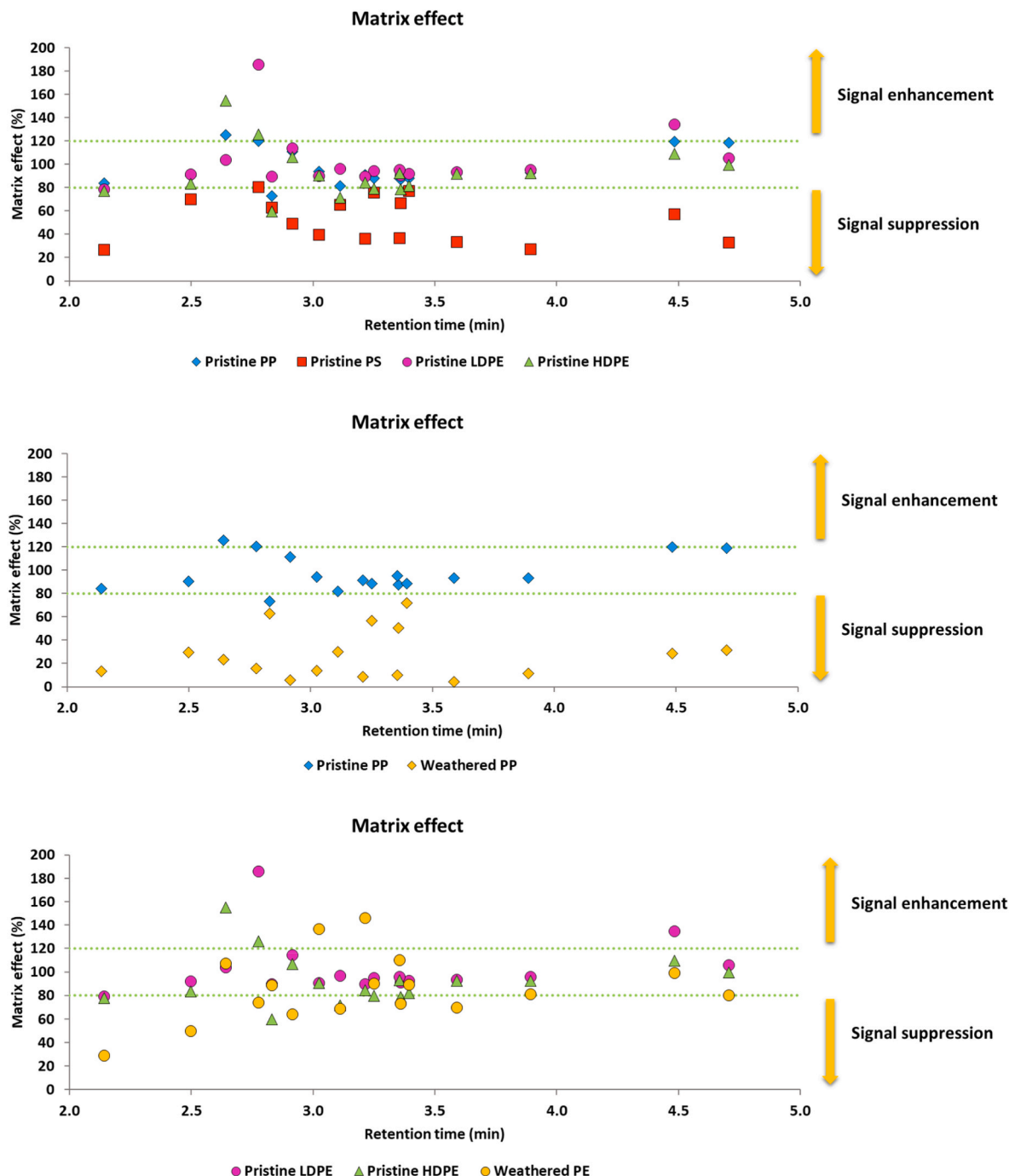


Fig. 5. Matrix effect of the different pristine and weathered MPs studied.

effect was calculated as the average of the ratio between the area of the analyte in the matrix-matched calibration and the area of the analyte in the solvent calibration, at each concentration level of the calibration curves. According to Matuszewski's method, negligible matrix effect takes place for values between 80 and 120 %, while higher matrix effect indicates a significant signal enhancement, and lower ones a significant signal suppression, being necessary to take this effect into account in both cases.

In this sense, matrix effect was assessed both in pristine and weathered MPs (see Fig. 5), to determine whether there are significant differences between the different polymeric matrices after the extraction. The evaluation of the matrix effect of pristine samples of PP, LDPE, and HDPE shows, overall, low matrix effect for most of the analytes (between 80 and 120 %), however, some of the analytes like trimethoprim or cyclophosphamide show a significant signal enhancement. In the case PS, on the other hand, there can be seen a significant signal suppression for all the analytes except for cyclophosphamide which has no significant matrix effect. The explanation to this difference could be found in the partial dissolution of the PS polymer, which despite being small, due to the limited amount of time in which plastic and solvent are in contact, could still be perceived. Nonetheless, no effect on the chromatographic separation nor in the detection could be observed over time.

Moving to weathered MPs and comparing them to their pristine counterparts, it can be easily observed that weathered MPs have a higher matrix effect, probably as a result of the partial dissolution of the plastic, which due to the effect of weathering is now more polar [35]. The comparison between pristine LDPE and HDPE and weathered PE, shows an overall higher matrix effect for the weathered MPs, although the differences between polymers is not excessively high. On the other hand, weathered PP shows a high signal suppression effect for all analytes, which is a strong contrast when compared to the pristine polymer, which has almost no matrix effect for most of the analytes except for trimethoprim, and cyclophosphamide which have a signal enhancement, and bisphenol F which has a signal suppression effect.

Overall, the evaluation of the matrix effect studies evidences the importance of assessing each type of polymer as a separate matrix, as there can be seen clear divergences between the different compositions, impeding the extrapolation of the validation done in one type of polymer to any kind of composition. Additionally, from the results shown, it is clear the necessity of carrying out matrix-matched calibration in all the studied polymers, as there are significant variations between the different calibration curves.

3.4. Analysis of real samples

In order to demonstrate the applicability of the proposed methodology, a total of ten environmental samples collected at Playa Grande beach (Tenerife, Canary Islands, Spain) were analysed (see Table S7 of the Supplementary Material). Due to the high abundance in the environment (i.e. beaches) of PE MP pellets [38–40], it was only possible to carry out the studies in this type of polymer; nevertheless, in future applications, more samples are expected to be treated. After proper characterization, 10 samples (identified as PE using ATR-FTIR) were cryomilled following the procedure previously described in the experimental section. After all of this, the extraction procedure was applied, and the reconstituted extracts injected in the UHPLC-MS/MS system. The evaluation of the 10 samples revealed that for most of them, none of the analytes were detected, with the exception of bisphenol A and caffeine which were found in 50 % and 10 % of the samples, respectively, mostly at concentrations below the LOQ of the method (10.8 ng/g for bisphenol A and 13.9 ng/g for caffeine). Overall, from the 10 samples studied, only one revealed the presence of bisphenol A at a concentration of 10.8 ± 3.4 ng/g. Bisphenol A is a plasticiser with a wide global production, that has numerous applications in the manufacturing of polycarbonate plastics and epoxy resins utilized in a variety of consumer

goods [41]. The effect of bisphenol A on human health has been linked to its potential endocrine disruptive action, altering molecular mechanisms, and, affecting bioavailability, which can cause important changes in cellular and physiological processes [42]. Furthermore, bisphenol A can potentially influence early-stage development by mimicking hormones like estradiol, being able to traverse both the placental and blood-brain barriers [43], and even accumulate on human placenta [44]. Additionally, as a result of its lipophilic nature, bisphenol A can permeate cell membranes and accumulate in fatty tissues [45]. Nevertheless, despite all of these evidence, further research on the effects BPA are still needed, in order to fully understand the risks associated with its use.

Overall, and as previously commented, research up to this date regarding the extraction of pollutants from MPs has been limited [12,14,30,31], with more publications being published over time, but it is in its initial steps as a result of the important challenges that face their determination (low recovery values, need of replicate analysis, proper assessment of matrix effects, among others). In this sense, most publications have focused on persistent organic pollutants. With regards to the most studied compounds, pollutants like PAHs or PCBs coped most of the research articles [12,14,30,31]. As reported in the review of Santana-Viera et al. [14] the predominant dataset focuses on PAHs and PCBs, measured in MPs collected from global open ocean and coastal regions; primarily on the Atlantic and North Pacific islands, North Pacific Ocean, and some select countries; with a notable absence of studies in areas with substantial human activity (e.g., Mediterranean Sea, Indian Ocean, etc.). On another review by Hong et al. [30], the main characteristics of the various MPs samples used in chemical analysis were discussed. As described in the publication, samples were predominantly collected from beaches, with no representation from bottom sediments and limited representation of sea surface MPs. With regards to the shape of the MPs, most studies focused mainly on pellets or fragments, with particle sizes ranging from mm to cm, with few examples of studies carried out on particles lower than 1 mm. As for the amount of sample used, given the difficulties arising from the collection of significant quantities of MPs to carry out the studies, most publications worked with amounts ≤ 1 g. Finally, chemical classification of the different MPs was not assessed in the majority of the studies, nonetheless, those that carried out the classification revealed a prevalence of PE, PP, and PS MPs.

4. Conclusions

A simple and effective methodology for the extraction of 17 emerging organic pollutants from pristine (PP, LDPE, HDPE, and PS) and weathered (PP and PE) MPs has been developed, optimized, and validated. The proposed method was applied for the analysis of 10 samples collected in Playa Grande (Tenerife, Canary Islands, Spain) revealing the presence of bisphenol A and caffeine in concentrations lower than the LOQ of the method, except for one of the samples where bisphenol A was found at a concentration of 10.8 ± 3.4 ng/g. This study represents the first simultaneous evaluation of particle size, weathering and matrix effect on MPs. In this sense, results have revealed the importance of cryomilling MPs (as smaller particle sizes improve recovery values and homogenise samples), a factor that has been widely overlooked in previous publications. Additionally, due to the differences in the matrix effect observed for the different types of pristine polymers, it is clear that polymeric matrices must be treated individually, without extrapolating the validation carried out in one type of polymer to others. In the case of weathered MPs, the differences are even accentuated, as matrix effect is overall higher in them than in their pristine counterparts, highlighting the necessity of always carrying matrix effect and recovery studies in the environmental MPs samples, as their characteristics diverge from those of the pristine MPs.

CRediT authorship contribution statement

Gabriel Jiménez-Skrzypek: Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Rachele Lusiardi:** Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft. **Javier González-Sálamo:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Daura Vega-Moreno:** Conceptualization, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Javier Hernández-Borges:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Javier González-Sálamo and Gabriel Jiménez-Skrzypek reports financial support was provided by Canarian Agency for Research, Innovation and Information Society. Javier Hernández-Borges reports financial support was provided by Spain Ministry of Science and Innovation. Javier Hernández-Borges reports financial support was provided by The Azores-Madeira-Canary Islands Transnational Cooperation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

G.J.S. and J.G.S. thank ACISI for the FPI contract and the Viera y Clavijo program contract, respectively, developed at the University of La Laguna (85 % co-financed by the European Social Fund). The authors are grateful for the financial support of the Azores-Madeira-Canary Islands Transnational Cooperation Program for the “IMPLAMAC” project (reference number MAC2/1.1a/265) financed with FEDER funds and project PID2020-112867 GB-I00 financed by MCIN/AEI/10.13039/501100011033.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aca.2023.342071>.

References

- [1] Plastics_Europe, Plastics - the Facts 2023, 2023. <https://plasticseurope.org/knowledge-hub/plastics-the-fast-facts-2023/>. (Accessed 22 November 2023).
- [2] R. Geyer, J.R. Jambeck, K.L. Law, Production, use, and fate of all plastics ever made, *Sci. Adv.* 3 (2023), e1700782, <https://doi.org/10.1126/sciadv.1700782>.
- [3] J.R. Jambeck, R. Geyer, C. Wilcox, T.R. Siegler, M. Perryman, A. Andrady, et al., Plastic waste inputs from land into the ocean, *Science* 347 (2015) 768–771, <https://doi.org/10.1126/science.1260352>, 1979.
- [4] S.B. Borrelle, J. Ringma, K.L. Law, C.C. Monnahan, L. Lebreton, A. McGivern, et al., Predicted growth in plastic waste exceeds efforts to mitigate plastic pollution, *Science* 369 (2020) 1515–1518, <https://doi.org/10.1126/science.aba3656>, 1979.
- [5] X. Guo, J. Wang, The chemical behaviors of microplastics in marine environment: a review, *Mar. Pollut. Bull.* 142 (2019) 1–14, <https://doi.org/10.1016/j.marpolbul.2019.03.019>.
- [6] C.B. Crawford, B. Quinn, in: C.B. Crawford, B.B.T.-M.P. Quinn (Eds.), 5 - Microplastics, Standardisation and Spatial Distribution, Elsevier, 2017, pp. 101–130, <https://doi.org/10.1016/B978-0-12-809406-8.00005-0>.
- [7] M. Cole, P. Lindeque, C. Halsband, T.S. Galloway, Microplastics as contaminants in the marine environment: a review, *Mar. Pollut. Bull.* 62 (2011) 2588–2597, <https://doi.org/10.1016/j.marpolbul.2011.09.025>.
- [8] Y. Zhang, S. Kang, S. Allen, D. Allen, T. Gao, M. Sillanpää, Atmospheric microplastics: a review on current status and perspectives, *Earth Sci. Rev.* 203 (2020), <https://doi.org/10.1016/j.earscirev.2020.103118>.
- [9] J.-J. Guo, X.-P. Huang, L. Xiang, Y.-Z. Wang, Y.-W. Li, H. Li, et al., Source, migration and toxicology of microplastics in soil, *Environ. Int.* 137 (2020), <https://doi.org/10.1016/j.envint.2019.105263>.
- [10] K. Ugwu, A. Herrera, M. Gómez, Microplastics in marine biota: a review, *Mar. Pollut. Bull.* 169 (2021), 112540, <https://doi.org/10.1016/j.marpolbul.2021.112540>.
- [11] C. Hernández-Sánchez, J. González-Sálamo, C. Ortega-Zamora, S. Hernández-Borges, J. Microplastics, An emerging and challenging research field, *Curr. Anal. Chem.* 16 (2020) 1–8, <https://doi.org/10.2174/1573411016999201029194655>.
- [12] G. Jiménez-Skrzypek, C. Hernández-Sánchez, C. Ortega-Zamora, J. González-Sálamo, M.A. González-Curbelo, J. Hernández-Borges, Microplastic-adsorbed organic contaminants: analytical methods and occurrence, *TrAC, Trends Anal. Chem.* 136 (2021), 116186, <https://doi.org/10.1016/j.trac.2021.116186>.
- [13] L. Fu, J. Li, G. Wang, Y. Luan, W. Dai, Adsorption behavior of organic pollutants on microplastics, *Ecotoxicol. Environ. Saf.* 217 (2021), 112207, <https://doi.org/10.1016/j.ecoenv.2021.112207>.
- [14] S. Santana-Viera, S. Montesdeoca-Esponda, R. Guedes-Alonso, Z. Sosa-Ferrera, J. J. Santana-Rodríguez, Organic pollutants adsorbed on microplastics: analytical methodologies and occurrence in oceans, *Trends in Environmental Analytical Chemistry* 29 (2021), e00114, <https://doi.org/10.1016/j.teac.2021.e00114>.
- [15] F. Yu, C. Yang, Z. Zhu, X. Bai, J. Ma, Adsorption behavior of organic pollutants and metals on micro/nanoplastics in the aquatic environment, *Sci. Total Environ.* 694 (2019), <https://doi.org/10.1016/j.scitotenv.2019.133643>.
- [16] G. Jiménez-Skrzypek, C. Ortega-Zamora, J. González-Sálamo, C. Hernández-Sánchez, J. Hernández-Borges, The current role of chromatography in microplastic research: plastics chemical characterization and sorption of contaminants, *J. Chromatogr. Open* 1 (2021), 100001, <https://doi.org/10.1016/j.jcoa.2021.100001>.
- [17] S. Endo, A.A. Koelmans, in: H. Takada, H.K. Karapanagioti (Eds.), Sorption of Hydrophobic Organic Compounds to Plastics in the Marine Environment: Equilibrium BT - Hazardous Chemicals Associated with Plastics in the Marine Environment, Springer International Publishing, Cham, 2019, pp. 185–204, <https://doi.org/10.1007/978-94-007-698-11>.
- [18] J.P. Rodrigues, A.C. Duarte, J. Santos-Echeandía, T. Rocha-Santos, Significance of interactions between microplastics and POPs in the marine environment: a critical overview, *TrAC, Trends Anal. Chem.* 111 (2019) 252–260, <https://doi.org/10.1016/j.trac.2018.11.038>.
- [19] P.S. Tourinho, V. Kočí, S. Loureiro, C.A.M. van Gestel, Partitioning of chemical contaminants to microplastics: sorption mechanisms, environmental distribution and effects on toxicity and bioaccumulation, *Environ. Pollut.* 252 (2019) 1246–1256, <https://doi.org/10.1016/j.envpol.2019.06.030>, 1987.
- [20] S.C. George, S. Thomas, Transport phenomena through polymeric systems, *Prog. Polym. Sci.* 26 (2001) 985–1017, [https://doi.org/10.1016/S0079-6700\(00\)00036-8](https://doi.org/10.1016/S0079-6700(00)00036-8).
- [21] A. Menéndez-Pedraza, J. Jaumot, Interaction of environmental pollutants with microplastics: a critical review of sorption factors, bioaccumulation and ecotoxicological effects, *Toxics* 8 (2020), <https://doi.org/10.3390/toxics8020040>.
- [22] T. Atugoda, M. Vithanage, H. Wijesekara, N. Bolan, A.K. Sarmah, M.S. Bank, et al., Interactions between microplastics, pharmaceuticals and personal care products: implications for vector transport, *Environ. Int.* 149 (2021), 106367, <https://doi.org/10.1016/j.envint.2020.106367>.
- [23] N.B. Hartmann, S. Rist, J. Bodin, L.H.S. Jensen, S.N. Schmidt, P. Mayer, et al., Microplastics as vectors for environmental contaminants: exploring sorption, desorption, and transfer to biota, *Integrated Environ. Assess. Manag.* 13 (2017) 488–493, <https://doi.org/10.1002/ieam.1904>.
- [24] G. Caruso, Microplastics as vectors of contaminants, *Mar. Pollut. Bull.* 146 (2019) 921–924, <https://doi.org/10.1016/j.marpolbul.2019.07.052>.
- [25] D. Brennecke, B. Duarte, F. Paiva, I. Caçador, J. Canning-Clode, Microplastics as vector for heavy metal contamination from the marine environment, *Estuar. Coast Shelf Sci.* 178 (2016) 189–195, <https://doi.org/10.1016/j.ecss.2015.12.003>.
- [26] A.M. Gulizia, K. Patel, B. Philippa, C.A. Motti, L. van Herwerden, G. Vamvounis, Understanding plasticiser leaching from polystyrene microplastics, *Sci. Total Environ.* 857 (2023), 159099, <https://doi.org/10.1016/j.scitotenv.2022.159099>.
- [27] H. Luo, Y. Zhao, Y. Li, Y. Xiang, D. He, X. Pan, Aging of microplastics affects their surface properties, thermal decomposition, additives leaching and interactions in simulated fluids, *Sci. Total Environ.* 714 (2020), 136862, <https://doi.org/10.1016/j.scitotenv.2020.136862>.
- [28] H. Luo, Y. Li, Y. Zhao, Y. Xiang, D. He, X. Pan, Effects of accelerated aging on characteristics, leaching, and toxicity of commercial lead chromate pigmented microplastics, *Environ. Pollut.* 257 (2020), 113475, <https://doi.org/10.1016/j.envpol.2019.113475>.
- [29] H. Chen, Z. Zou, M. Tang, X. Yang, Y.F. Tsang, Polycarbonate microplastics induce oxidative stress in anaerobic digestion of waste activated sludge by leaching bisphenol A, *J. Hazard Mater.* 443 (2023), 130158, <https://doi.org/10.1016/j.jhazmat.2022.130158>.
- [30] S.H. Hong, W.J. Shim, L. Hong, Methods of analysing chemicals associated with microplastics: a review, *Anal. Methods* 9 (2017) 1361–1368, <https://doi.org/10.1039/c6ay02971j>.

- [31] G. Jiménez-Skrzypek, J. González-Sálamo, J. Hernández-Borges, Chapter 17 - application of liquid chromatography in studies of microplastics, in: S. Fanali, B. Chankvetadze, P.R. Haddad, C.F. Poole, M.-L. Riekkola (Eds.), *Liquid Chromatography*, third ed., vol. 2, Elsevier, 2023, pp. 579–605, <https://doi.org/10.1016/B978-0-323-99969-4.00017-6>.
- [32] G. Jiménez-Skrzypek, O.M. Hernández-Expósito, J. Hernández-Borges, J. González-Sálamo, Sorption of levonorgestrel on polyethylene, polystyrene and polypropylene microplastics, *Chemosphere* 335 (2023), 139042, <https://doi.org/10.1016/j.chemosphere.2023.139042>.
- [33] B. Smith, The infrared spectra of polymers III: hydrocarbon polymers, *Spectroscopy* 36 (2021) 22–25.
- [34] B. Smith, The infrared spectra of polymers II: polyethylene, *Spectroscopy* 36 (2021) 24–29.
- [35] V. Fernández-González, J.M. Andrade-Garda, P. López-Mahía, S. Muniategui-Lorenzo, Impact of weathering on the chemical identification of microplastics from usual packaging polymers in the marine environment, *Anal. Chim. Acta* 1142 (2021) 179–188, <https://doi.org/10.1016/j.aca.2020.11.002>.
- [36] S. Venkatram, C. Kim, A. Chandrasekaran, R. Ramprasad, Critical assessment of the Hildebrand and Hansen solubility parameters for polymers, *J. Chem. Inf. Model.* 59 (2019) 4188–4194, <https://doi.org/10.1021/acs.jcim.9b00656>.
- [37] B.K. Matuszewski, M.L. Constanzer, C.M. Chavez-Eng, Strategies for the assessment of matrix effect in quantitative bioanalytical methods based on HPLC–MS/MS, *Anal. Chem.* 75 (2003) 3019–3030, <https://doi.org/10.1021/ac020361s>.
- [38] N. Digka, C. Tsangaris, H. Kaberi, A. Adamopoulou, C. Zeri, Microplastic abundance and polymer types in a mediterranean environment, in: M. Cocca, E. Di Pace, M.E. Errico, G. Gentile, A. Montarsolo, R. Mossotti (Eds.), *Proceedings of the International Conference on Microplastic Pollution in the Mediterranean Sea*, Springer International Publishing, Cham, 2018, pp. 17–24.
- [39] S.-K. Kim, J.-S. Kim, H. Lee, H.-J. Lee, Abundance and characteristics of microplastics in soils with different agricultural practices: importance of sources with internal origin and environmental fate, *J. Hazard Mater.* 403 (2021), 123997, <https://doi.org/10.1016/j.jhazmat.2020.123997>.
- [40] M. Tiwari, T.D. Rathod, P.Y. Ajmal, R.C. Bhangare, S.K. Sahu, Distribution and characterization of microplastics in beach sand from three different Indian coastal environments, *Mar. Pollut. Bull.* 140 (2019) 262–273, <https://doi.org/10.1016/j.marpolbul.2019.01.055>.
- [41] L.N. Vandenberg, R. Hauser, M. Marcus, N. Olea, W.V. Welshons, Human exposure to bisphenol A (BPA), *Reprod. Toxicol.* 24 (2007) 139–177, <https://doi.org/10.1016/j.reprotox.2007.07.010>.
- [42] A. Martínez-Ibarra, L.D. Martínez-Razo, K. MacDonald-Ramos, M. Morales-Pacheco, E.R. Vázquez-Martínez, M. López-López, et al., Multisystemic alterations in humans induced by bisphenol A and phthalates: experimental, epidemiological and clinical studies reveal the need to change health policies, *Environ. Pollut.* 271 (2021), 116380, <https://doi.org/10.1016/j.envpol.2020.116380>.
- [43] A. Abraham, P. Chakraborty, A Review on Sources and Health Impacts of Bisphenol A, vol. 35, 2020, pp. 201–210, <https://doi.org/10.1515/reveh-2019-0034>.
- [44] S. Gilbert, W. Werner, H. Hartmut, T.C. E. P. Martin, C. Ibrahim, Parent bisphenol A accumulation in the human maternal-fetal-placental unit, *Environ. Health Perspect.* 110 (2002) A703–A707, <https://doi.org/10.1289/ehp.110-1241091>.
- [45] M.F. Fernandez, J.P. Arrebola, J. Taoufik, A. Navalón, O. Ballesteros, R. Pulgar, et al., Bisphenol-A and chlorinated derivatives in adipose tissue of women, *Reprod. Toxicol.* 24 (2007) 259–264, <https://doi.org/10.1016/j.reprotox.2007.06.007>.